Lemma 1.4: Let M be a paracompact Hausdorff space and $\{U_\alpha\}_{\alpha\in A}$ *any* open cover. Then there exists a locally finite open cover $\{V_\alpha\}_{\alpha\in A}$ such that $\overline{V_\alpha} \subseteq U_\alpha$ for all α.

Proof:

Our first step is showing that paracompact and Hausdorff implies regular (i.e. T_3).

Let $A \subseteq M$ be a closed set and suppose $p \notin A$. Then since M is Hausdorff, for all $q \in A$ we can pick an open set $V_q \subseteq M$ such that $p \notin \overline{V_q}$. Next, note that $\{M - A\} \cup \{V_q\}_{q\in A}$ is an open cover of M. So by paracompactness, we can take a locally finite refinement $\{W_\beta\}_{\beta\in B}$ covering M.

Let B' be the collection of $\beta \in B$ such that $W_\beta \cap A \neq \emptyset$. Since we know that $M = \bigcup_{\beta\in B} W_\beta$, we can thus conclude that $A \subseteq \bigcup_{\beta\in B'} W_\beta$. At the same time though, because $\{W_\beta\}_{\beta\in B'}$ is locally finite, we can say that:

$\overline{\bigcup_{\beta\in B'} W_\beta} = \bigcup_{\beta\in B'} \overline{W_\beta} \subseteq \bigcup_{q\in A} \overline{V_q}$ where the latter set doesn't contain p.

So, the open sets $U := \bigcup_{\beta\in B'} W_\beta$ and $V := M - \overline{U}$ separate p and the closed set A.

Next, let $\mathcal{A}$ be the collection of open subsets of M whose closures are contained in a set U_α. Because M is regular, we know that $\mathcal{A}$ covers all of M. So by paracompactness, we can take a locally finite refinement $\{W_\beta\}_{\beta\in B}$ of $\mathcal{A}$ covering M.

Define a function $f : B \to A$ such that $\overline{W_\beta} \subseteq U_{f(\beta)}$ for all β. Then for every α define $V_\alpha := \bigcup_{\beta\in f^{-1}(\{\alpha\})} W_\beta$. Clearly $\bigcup_{\alpha\in A} V_\alpha = \bigcup_{\beta\in B} W_\beta = M$. Also note that because the W_β are locally finite:

$$\overline{V_\alpha} = \overline{\bigcup_{\beta\in f^{-1}(\{\alpha\})} W_\beta} = \bigcup_{\beta\in f^{-1}(\{\alpha\})} \overline{W_\beta} \subseteq U_\alpha. \blacksquare$$

Theorem 2.23: Let M be a paracompact Hausdorff space and let $\mathcal{U} = \{U_\alpha\}_{\alpha\in A}$ be *any* open cover. Then there is a continuous partition of unity $\{f_\alpha : M \to [0,1]\}_{\alpha\in A}$ subordinate to $\mathcal{U}$ such that $\operatorname{supp}(f_\alpha) \subseteq U_\alpha$ for all $\alpha \in A$.

Proof:

We know from before that paracompact and Hausorff implies regular. As it turns out, we can also use an identical argument (except that we replace p with a disjoint closed set B from A) in order to prove that paracompact and regular implies normal (i.e. T_4). This is useful because we can apply Urysohn's lemma to any normal space.

By lemma 1.4, we can find two locally finite open covers $\{V_\alpha\}_{\alpha\in A}$ and $\{W_\alpha\}_{\alpha\in A}$ such that $\overline{W_\alpha} \subseteq V_\alpha \subseteq \overline{V_\alpha} \subseteq U_\alpha$ for all α. And then by Urysohn's lemma we can for every $\alpha \in A$ find a function $g \in C(M, [0,1])$ such that $g \equiv 1$ on $\overline{W_\alpha}$ and $g \equiv 0$ on $M - V_\alpha$.

But now note that as $\operatorname{supp}(g_\alpha) \subseteq \overline{V_\alpha}$ and $\{\overline{V_\alpha}\}_{\alpha\in A}$ is locally finite, we know that every $p \in M$ has a neighborhood on which only finitely many g_α are nonzero. So, we can define a continuous function $h = \sum_{\alpha\in A} g_\alpha$. And then we define $f_\alpha = g_\alpha/h$ for all α. This satisfies what we want. $\blacksquare$

Here is a proposition about locally finite collections of sets.

Exercise 1.14: Suppose $\mathcal{X}$ is a locally finite collection of subsets of a topological space M.

(a) The collection $\{\overline{X} : X \in \mathcal{X}\}$ is also locally finite.

Proof:
Let $x \in M$ and let $U \subseteq M$ be an open set containing x that intersects only the elements of a finite subset $\mathcal{Y}$ of $\mathcal{X}$. Now we want to show that if $X \in \mathcal{X}$ satisfies $\overline{X} \cap U \neq \emptyset$, then $X \in \mathcal{Y}$. Luckily, if $\overline{X} \cap U \neq \emptyset$ but $X \cap U = \emptyset$ so that $X \notin \mathcal{Y}$, then it must be that any $x \in \overline{X} \cap U$ is an accumulation point of X. However, that is immediately contradicted by the fact that U is a neighborhood of x which does not intercept X anywhere. ■

(b) $\overline{\bigcup_{X\in\mathcal{X}} X} = \bigcup_{X\in\mathcal{X}} \overline{X}$.

Proof:
It is always true that $\bigcup_{X\in\mathcal{X}} \overline{X} \subseteq \overline{\bigcup_{X\in\mathcal{X}} X}$.

Meanwhile, to show the other inclusion, suppose x is an accumulation point of $\bigcup_{X\in\mathcal{X}} X$ but not in any individual $\overline{X}$. Then it must be the case that every neighborhood of x intercepts some set in $\mathcal{X}$, yet it must also be the case that for every $X \in \mathcal{X}$ there is a neighborhood of x with doesn't intercept X. The only way this is possible is if every neighborhood intercepts infinitely many $X \in \mathcal{X}$.

Let N be any neighborhood of x. Then we can construct an infinite sequence of distinct sets $(X_n)_{n\in\mathbb{N}}$ in $\mathcal{X}$ intercepting N as follows.

For the ease of notation set $U_0 = M$. Now at step n, choose any $X_n \in \mathcal{X}$ that intersects $N \cap \bigcap_{k=0}^{n-1} U_k$. Note that we can do this since a finite intersection of neighborhoods of x is still a neighborhood of x. But now we can choose some neighborhood U_n of x such that $X_n \cap U_n = \emptyset$. And now repeat this reasoning.

It's clear that all the chosen X_n intercept $\bigcap_{k=0}^{n-1} U_k \subseteq N$. Additionally, all the chosen X_n are distinct since $X_n \cap \bigcap_{k=0}^{N} U_k \subseteq X_n \cap U_n = \emptyset$ for all $N \geq n$.

But that contradicts that $\mathcal{X}$ is locally finite. ■